

Frozen element embeddings versus categorical species in linear ACE: distillation benchmarks on CrMnFeCoNi

ACEpotentials.jl — 14 September 2026 — accompanies pull requests from `fix/basis-ed-performance`, `pr/jax-port`, `pr/element-embeddings`

Summary

A linear ACE model whose species dependence enters through a **frozen element embedding** (Darby-style tensor reduction, channel folded into the radial index) was benchmarked against the standard **categorical** model on a five-component alloy, with labels distilled from MACE-MH-1. At matched polynomial degree the embedded model costs 7–17% on force RMSE for 3–6× fewer parameters; given one more degree — which its smaller design matrix makes affordable — it **matches** the categorical model at 2.4× fewer parameters (0.0767 vs 0.0763 eV/Å). The best absolute result is categorical degree 10 on 4 000 structures, 0.0641 eV/Å and 2.0 meV/atom, fitted from a 151 GB design matrix. Learning curves show the categorical model is data-limited and the embedded model basis-limited, which is why the trade runs the way it does. Two surprises: the embedding’s **values** do not matter at $S = 5$, and the Julia evaluator ran the embedded model **slower** than the categorical one — now profiled and fixed. Recommendations, including code changes in flight, close the report.

Set-up

System and labels. Rattled and strained fcc CrMnFeCoNi cells (32–48 atoms, random equiatomic occupancy, no short-range order), generated with bounded uniform perturbations (lattice $\pm 3\%$, shear 0.02, rattle 0.02–0.10 Å, minimum separation 2.0 Å). Energies, forces and virials from MACE-MH-1 on an RTX A4500 (4 000 structures in 4.6 min). An earlier 6%-Gaussian generator produced crushed cells (36% compression, forces above 20 eV/Å) carrying 77% of the squared test error; bounding it halved the error at fixed basis and budget. MACE-MP-0-small was rejected as teacher: a difference-fit showed its disagreement with MH-1 to be rough, not a smooth offset.

Models. Order 3, ACE1 heuristics, uniform cutoff 6.25 Å. Categorical `ace1_model` with $S = 5$ elements; embedded `ace_embedding_model` with per-order widths $d_\nu = \min(d_{\max}, \binom{S+\nu-1}{\nu})$: **lossless** [5, 15, 35] (the full species-tensor width at every order; not a reparameterisation of the categorical model — it has strictly fewer functions, 475 against 1 348 at degree 6, because degree truncation treats the folded species index differently), $d_{\max} = 16$ ([5, 15, 16], order 3 cut to 16 of 35 directions) and $d_{\max} = 8$. Embedding table: MACE-MH-1’s 512-channel `node_embedding` reduced by PCA of the five rows (Gram preserved exactly; generic orthonormal mixing for $d > S$). Parameters are $n_B \times S$ (a readout per central species).

Fitting. Held-out 80/20 split; algebraic smoothness prior $p = 4$; weights from `acefit!` defaults; factor-once Tikhonov (QR of A , SVD of R) with λ swept per model and the best held-out force RMSE reported. Assembly on 12 workers; the degree-10 categorical matrix (404 144 × 46 885, 151 GB) was factorised in place on a 376 GB node.

branch (PR)	based on	contents
<code>fix/basis-ed-performance</code>	<code>main</code>	assembly and forward-evaluation speed-ups; profiling findings; 10 commits
<code>pr/jax-port</code>	<code>fix/basis-ed-performance</code>	JAX evaluator port: <code>acejax</code> , exporter, tests + CI divergence gate, LAMMPS, benchmarks, plan; 7 commits, no changes under <code>src/</code>
<code>pr/element-embeddings</code>	<code>pr/jax-port</code>	frozen element embeddings, factorised radial export, distillation benchmarks, FS/VarPro spikes, this report; 5 commits

Table 1: The three pull requests. Fine-grained history (146 commits) is on branch `jax-eval`; the PR branches are squashed into logical commits.

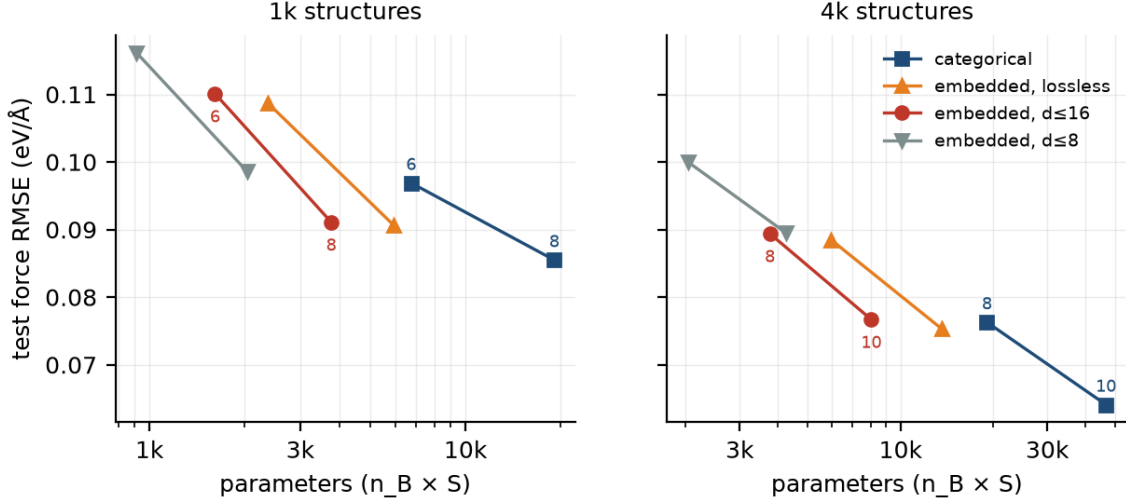


Figure 1: Test force RMSE against parameter count. Point labels are the polynomial degree. On the 4k set the degree-10 embedded $d \leq 16$ model (8 045 parameters) sits at the accuracy of degree-8 categorical (19 120).

Results

model	n_B	params	1k test F	4k test F	4k test E	rel. to cat.
deg 6 categorical	1 348	6 740	0.0969	—	—	—
deg 6 embedded $d \leq 16$	323	1 615	0.1101	—	—	+14%, 4.2× fewer (1k)
deg 8 categorical	3 824	19 120	0.0855	0.0763	0.0024	—
deg 8 embedded lossless	1 190	5 950	0.0906	0.0885	0.0025	+16%, 3.2× fewer
deg 8 embedded $d \leq 16$	753	3 765	0.0911	0.0894	0.0025	+17%, 5.1× fewer
deg 8 embedded $d \leq 8$	408	2 040	0.0986	0.1000	0.0027	+31%, 9.4× fewer
deg 10 categorical	9 327	46 635	—	0.0641	0.0020	—
deg 10 embedded lossless	2 730	13 650	—	0.0753	0.0022	+17%, 3.4× fewer
deg 10 embedded $d \leq 16$	1 609	8 045	—	0.0767	0.0022	+20%, 5.8× fewer
deg 10 embedded $d \leq 8$	850	4 250	—	0.0895	0.0025	+40%, 11× fewer

Table 2: Held-out force RMSE (eV/Å) and energy RMSE (eV/atom) against MH-1 labels; 1k = 800/200 split, 4k = 3 200/800. “rel. to cat.” compares with the categorical model at the same degree on 4k.

Three things follow from Table 2 and Figure 1. First, at matched degree the embedding costs 14–20% on forces (energies are unaffected: 2.2–2.7 meV/atom throughout), for 3–6× fewer parameters. Second, **lossless and $d \leq 16$ are indistinguishable at every degree** — truncating order 3 from 35 to 16 species directions costs nothing measurable at $S = 5$ — whereas $d \leq 8$, which also truncates order 2, costs about one degree. Third, per basis function the embedded model wins at every point: degree-10 $d \leq 16$ matches degree-8 categorical at 2.4× fewer parameters, and it was affordable (26 GB) where the categorical alternative needed a 376 GB node.

The learning curves (Figure 2) explain the table. Both degree-6 models saturate — train and test meet — by 400 structures; the embedded model gets there by 200 and is the better model below that. At degree 8 the categorical model still has slope at 800 and duly gains 11% from 4× the data, while the embedded models gain 2%. **The embedded basis wants degree, not data**; the categorical basis wants data. This is the mechanism behind “accurate per structure vs accurate per parameter”.

The embedding’s values do not matter at $S = 5$

A 2×2 of embedding source (MP-0-small, MH-1) × reduction (first- d column truncation, PCA) at degree 6 gave test F within 0.001 eV/Å at $d \leq 16$ and 0.005 at $d \leq 8$, although the truncated tables’ element-similarity structure correlates only 0.5–0.7 with the full tables’. At $S = 5$ the frozen embedding acts as a generic tensor reduction, consistent with Darby et al.’s random-projection result — and with the tables themselves: the full 89×128 MP-0 table has rank 89 with a slow, featureless singular-value decay (64 components for 98% of the energy), i.e. no low-rank “chemical” structure to exploit; transferred chemistry can only show once $d_{\max} < S$ truncates order 1 itself. PCA remains the right construction (it was also what exposed that a DCT mixing frame spans only 9/15 and 13/35 of the species tensors — a genericity check now in the tests).

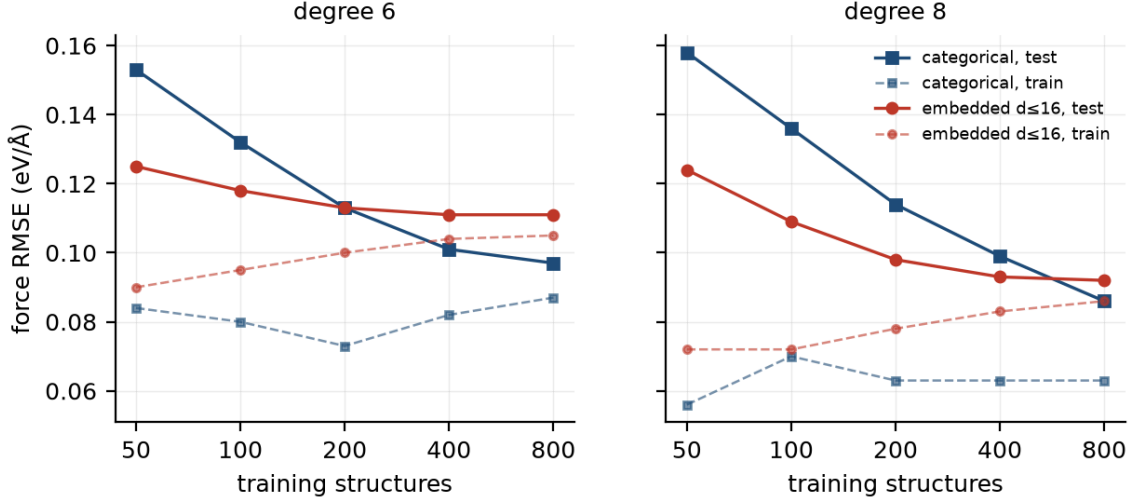


Figure 2: Learning curves on the 1k set (test solid, train dashed). Degree 6 is basis-limited for both models; at degree 8 the categorical model is still data-limited (gap 0.022 at $N = 800$) while the embedded one has converged.

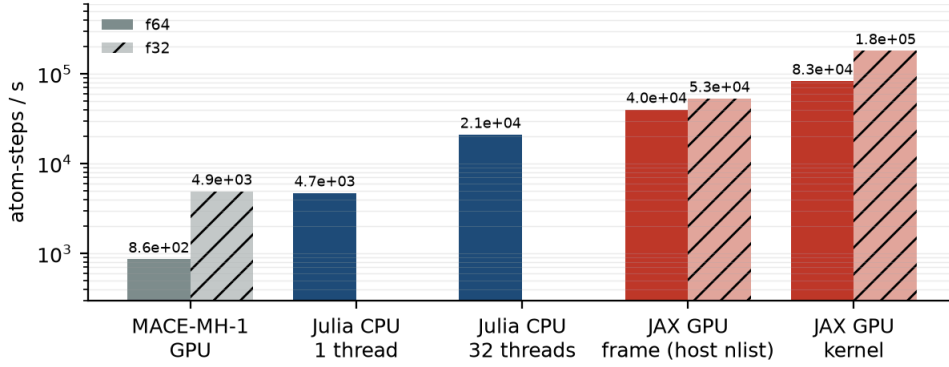


Figure 3: Throughput of the degree-6 $d \leq 16$ student on one host (lestrade: 32 cores, RTX 4000 Ada), 256–384-atom cells, f64 (solid) and f32 (hatched). The JAX kernel runs 32–38× the teacher in f32 and about 100× in f64; a single Julia core matches the teacher on a GPU. Julia is f64 only. Neither card is a production GPU: fp64 runs at 1/64 of fp32 on both the RTX 4000 Ada and the A4500. The f32 bars use XLA’s defaults, i.e. TF32 matmuls on these cards (a 1.2×10^{-3} descriptor error unless `jax_default_matmul_precision=highest` is set, which acejax’s production entry points do); the f64 bars are exact.

Evaluation cost

The JAX evaluator’s GPU kernel runs 32–38× MACE-MH-1’s in f32 at 40–1 300 atoms (6–11× end-to-end with the host neighbour list) and $7.4\text{--}8.3 \times 10^4$ atom-steps/s in f64 — about 100× MH-1’s f64 rate of $7.6\text{--}8.6 \times 10^2$, because the ACE kernel is gather-bound and barely notices the card’s 1/64 fp64 rate while MACE’s dense matmuls do; the same holds on the A4500 (4.8×10^4 f64 at 40 atoms). MH-1 does not fit a 3 000-atom cell on 20 GB in either precision.

None of the MACE numbers in this report use fused equivariant kernels, and they cannot on this stack.

Checked directly: the mace-jax bundles run the plain `e3nn-jax` path (`equivariance_config: null`); re-loading the same weights with the `cueq` backend enabled routes through `cuequivariance_jax`, but every mace-jax adapter except one calls it with `method='naive'` — the pure-JAX reference implementation, which XLA compiles exactly as it compiles `e3nn-jax` — so the HLO contains no custom kernels and the “accelerated” model is 2–20% **slower** (layout transposes), at 4×10^{-9} eV/Å agreement in f64. The one fused kernel (`uniform_1d`, channel-wise tensor product) is reached only with `conv_fusion=True` and needs the CUDA op package and Triton, neither of which the environment had. The 2–5× that `cuEquivariance` is reported to give MACE is a MACE-torch result. So the speed-ups here are **against the JAX MACE stack as shipped**; a like-for-like accelerated comparison would be against MACE-torch + `cuEquivariance`, a separate harness, and is the outstanding measurement. The Julia evaluator, however, ran the **embedded** model 1.8× slower than the categorical one despite 4× fewer basis functions, and both far below a well-written CPU kernel. Profiling found the cause: the splined radial tables are LEN wide (224 columns per edge for the embedded model), yet because the embedding weights make every R_{nl} a scalar multiple of $P_{n'}$, only 6–8 columns are distinct — the evaluator computes them all, as `SVector{LEN, Dual}`. A prototype exploiting the factorisation ($R_{nl} = c_{n,\text{pair}} P_{n'}$, the structure acejax already uses) with an allocation-free site loop is exact to 10^{-12} and 12–18× faster on

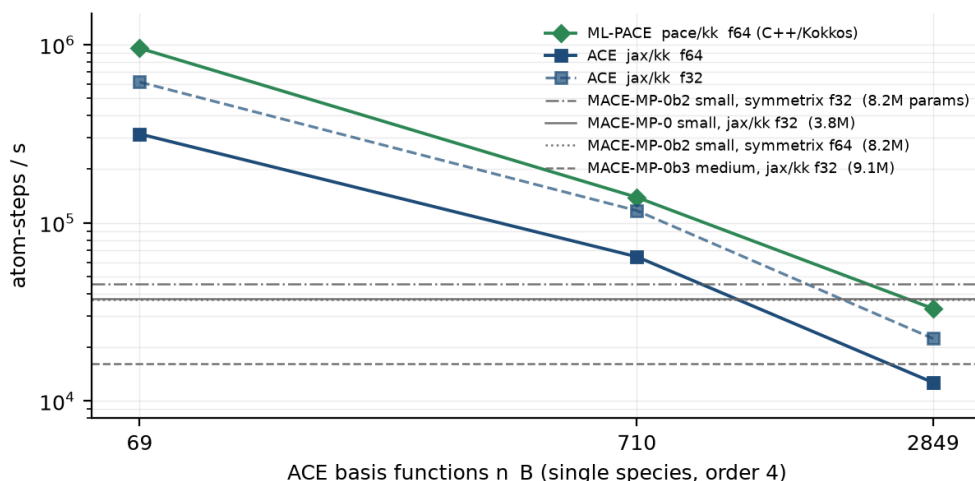


Figure 4: LAMMPS throughput at 1 728 Si atoms, one rank, one RTX A4500 (moriarty). ACE through pair_style jax/kk at three basis sizes (order 4, lmax 4–5, cutoff 6 Å), ML-PACE pace/kk at matched basis shape (78 / 693 / 2 874 functions, 0.9–13% off ours), and three MACE foundation models through jax/kk (lammps-jax) or symmetrix (hand-written Kokkos). Every engine was gated against the others before timing: ACE jax/kk vs Python 1e-13 eV/Å; symmetrix vs jax/kk on the same MACE checkpoint 4e-5 eV/Å.

four cores, above the JAX fp64 GPU kernel. On the JAX side the one known size-dependent cost — the reverse-mode scatter behind the edge gather, which degraded the Apple-Silicon MD path 2.3× between 216 and 1 728 atoms — has a measured cure (an algebraically identical one-hot matmul, 5.1× end-to-end with bit-identical trajectories) that is in acejax as edge_a_kind="matmul" but has not yet been calibrated on the A4500, where the kernel drops 2× between ≈330 and ≈1 100 atoms (Phase 15).

In LAMMPS: against ML-PACE and MACE

Figure 4 puts the JAX evaluator in its production setting. Against **ML-PACE** at matched basis, jax/kk is 2.1–3.0× slower in f64 and 1.2–1.5× in f32, roughly flat across basis size; the gap is not in the kernel — in Python the same model matches pace at matched basis — but in the plugin path (padding to a fixed edge capacity, ghost atoms, the comm mode; measured 3× at the large basis, 1.05× at the small). Against **MACE**, the 69- and 710-function ACE models run 16× and 3× MACE-MP-0 small through the **same** pair style; the 2 849-function model runs at 0.6×. **That last figure is the plugin path, not the model**: called from Python on the same card the same 2 849-function ACE does 5.5×10⁴ atom-steps/s in f32 — 1.5× MACE-in-LAMMPS — and jax/kk retains only 0.41 of it (0.34 in f64), against 0.55–0.63 for the two smaller bases. The loss is concentrated where the per-atom stages dominate (A2B 54%, AA 38% of site_basis at this shape) because our bundle pads the atom axis 3.7× (locals + ghosts + margin) and evaluates those stages over every slot, whereas the lammps-jax MACE bundle runs in comm mode with - -owned-rows and does not pay for ghosts. Two further contributions are card-scale rather than harness: a gather-bound kernel is bandwidth-limited on a 20 GB A4500 where MACE’s dense 128-channel matmuls are what the card does best, and the reverse-mode scatter behind the edge gather (Phase 15) is uncalibrated here. The fixes are known and cheap: owned-rows semantics for the ACE bundle, the Phase 15 calibration, and a MACE-in-Python point so both sides carry a retention figure. Two consequences for this report: the 5-element embedded models above sit at 323–1 609 functions, i.e. between the middle and large points here, so their LAMMPS throughput is expected in the 2–6×10⁴ range rather than the 10⁵ the isolated kernel reaches; and the same jax/kk engine runs MACE-MP-0b3 medium 1.4× **faster** than symmetrix does, so the JAX export path is not itself the handicap.

Stage 2 proposal, for discussion

The plan (docs/plans/jax_ace_port_plan.md, Stage 2) has been revised against these results. Its shape, and the decisions it asks for:

2A — linear fit at scale: an open choice. Two routes. (i) **Stay in Julia**: assembly and solve where the model, the priors, the parameter-gradient rules and the data path already live — a single source of truth — with **phase 18** (row-block assembly feeding TSQR, which still yields R for BLR, or ScaLAPACK on the same operator per the gap_fit recipe; LSQR beyond the R -factor floor) for what no single node holds. (ii) **Port assembly to JAX**: the design matrix built on the GPU through the same differentiable descriptor the evaluator uses, so fitter and evaluator cannot diverge, with the solve in whichever runtime holds A . The case for (ii) was strongest when Julia assembly was 100× slower than its arithmetic warranted; the performance branch removed that premise — the 151 GB degree-10 matrix now

assembles in 25 min on CPU, and basis pre-training (2B) needs nothing Julia lacks. What remains for (ii) is real but smaller: GPU bandwidth ($\approx 10\times$ on a gather-bound assembly, not $100\times$), the shared descriptor, and MACE’s own move to JAX. Sizing either way: at 4 000 structures the categorical degree-12 matrix is 330 GB and needs phase 18 on any route; the embedded one is 50 GB. The phase-18 spike settled the solver: ScaLAPACK.jl is dead (Julia-0.4 code) but the library works from SCALAPACK32_jll + MPI.jl; a 26-line TSQR over MPI.jl is the best direct method on ACEfit’s row-block layout (agreement with LAPACK to $1e-12$ at cond $1e21$, $13\times$ faster than ScaLAPACK on that layout), with the $n \times n$ R factor as its memory floor (162 GB at embedded degree 20); LSQR’s damp is QR’s λ for $\lambda \geq 10^{-3}$ and is the only route beyond that floor; GPU QR (JAX/lineax) is exact but only $1-2\times$ a 32-thread CPU at the sizes that fit 20 GB and OOMs at the degree-10 shape. A third route — compiling the Julia model to XLA with Reactant, which would make the language question moot — remains blocked: the standard ETACE path traces on a CUDA host except for composite (struct-of-array) inputs to KernelAbstractions kernels (EnzymeAD/Reactant.jl#3267), a data-layout change in EquivariantTensors rather than a compiler fix. Rust and PyTorch were assessed and set aside; the plan records the reasoning. *Decision sought: (i) or (ii) — or (i) now with (ii) held open, using the $S = 25$ gate experiment (a ≈ 150 GB assembly) as the test of whether the Julia route suffices at the scale that matters.*

2B — basis pre-training, then convex fit. Everything non-linear (radial mixing W_{nlq} , the element embedding E , Finnis–Sinclair densities as linear combinations of the order-1 basis) is learned once on a subset by Kaufman variable projection — the gradient at fixed c^* , which the existing grad_params / pullback_2_mixed / energy_forces_virial rrules already provide in Julia — then frozen, and the coefficients are fitted convexly at scale. The spike passed its gate: learned densities beat hand-chosen ones with fewer columns and transfer frozen to a disjoint split. This is the GRACE-FS / PACE function class split into two stages; pacemaker, gracemaker and MACE all learn their radials, and ACEpotentials’ fixed heuristic radials are the outlier. *Decision sought: make pre-trained radials and densities the default path rather than an option; two Julia items to build (a structured $W_{nlq}(E)$ and the FS term at the hook already marked in ace.jl).*

2C — the convexity demonstration against GRACE-FS. Neither pacemaker (linear, categorical) nor gracemaker (embedded, gradient-descent) fits a tensor-reduced ACE convexly; that is the differentiator, and it has to be shown on the data the competition used — HEA25S, teacher GRACE-2L-OMAT (public), their extended-distillation recipe. Five measurable benefits: calibrated UQ for free (BLR, and Swinburne–Perez POPS for the misspecified regime a distilled student is always in) against a GRACE-FS ensemble; exact D-optimal data selection (error vs N , selected vs random); one-shot deterministic fits with closed-form λ ; delta-learning on DFT as one more linear solve against GRACE’s fine-tuning curve; and hard physical constraints as convex QPs (equilibrium, elastic constants, phase ordering, repulsion). *Gate first: accuracy at matched MD cost at $S = 25$ on a 5 000-structure subset (≈ 150 GB, fits the 376 GB node today) — the first regime where $d_{\max} < S$ and the embedding’s values can matter. If the linear model does not reach GRACE-FS accuracy there, items 1–5 are wins on a worse model. Decision sought: approve the gate experiment as the next substantial run, ahead of phase 18.*

Upstream items. A latent correctness bug in EquivariantTensors, found by the embedding spike and not yet reported upstream: sparse_equivariant_tensor (the singular form, which ACEpotentials.Models._generate_ace_model calls) silently returns a **non-equivariant** basis when mb_spec is not grouped by correlation order — SparseSymmProd re-sorts the spec but the A2Bmaps columns are not permuted to match; measured rotation error 0.04–0.86 for ungrouped specs against 5×10^{-16} grouped. ACEpotentials is safe today only because its spec generator happens to emit order-grouped specs. The plural sparse_equivariant_tensors already guards against this; a fix with a regression test is on jameskermode/EquivariantTensors.jl:fix-spec-ordering (commit 31f8f13) and needs a PR. Then: multi-tangent pushforward! for PooledSparseProduct and SparseSymmProd in EquivariantTensors (replacing the three interim kernels in basis_ed.jl; ET’s own commented-out _p_fwd is most of it); SparseSymmProdDAG for the CPU evaluator; ACEfit’s per-task serialisation, GC.gc() and Array(A).

Recommendations

- Use the embedded model where parameters, memory or many elements are the constraint, and give it one more degree than the categorical model would get.** Degree-10 $d \leq 16$ is the current best trade (0.0767 eV/Å, 8 045 parameters, 26 GB matrix on 4k). Where accuracy per structure is all that matters and a large node is available, categorical degree 10 (0.0641) still wins; categorical degree 12 (330 GB) needs the distributed assembly of Stage 2 phase 18, embedded degree 12 (≈ 50 GB) does not.
- Keep $d_{\max} = 16$; do not use $d_{\max} = 8$** (it truncates order 2 and costs a full degree).
- Treat the embedding source as immaterial at $S \leq d_{\max}$,** and test the transferred-chemistry claim only at $S = 10-25$, where the GRACE-FS distillation on HEA25 is the comparison to beat.

4. **Code changes, as three stacked pull requests against main** (branches listed in Table 1). (a) **fix/basis-ed-performance**: the design-matrix path — `evaluate_basis_ed` replaced by a type-stable pushforward Jacobian and the boxed accumulation in `energy_forces_virial_basis` removed — 174–218× per structure, exact to 10^{-12} ; and the forward path — factorised radial spline tables with a plain-Float64 kernel, an allocation-free site loop with $w_{AA} = A2B^T W_B$ folding, neighbour-list reuse — 9–12× on four threads, exact to 10^{-12} ; 1 300 existing tests plus 471 new ones pass. The 151 GB degree-10 assembly above took 25 min on it instead of a projected 14 h. Its three pushforward kernels reimplement `EquivariantTensors`’ product structure and should move upstream as multi-tangent pushforward! methods (a follow-up PR to `EquivariantTensors`); the ACEfit follow-ups (per-task model serialisation, `GC.gc()` per task, the `Array(A)` copy) are listed in the PR text. (b) **pr/jax-port**: the JAX evaluator, exporter, LAMMPS bundle, benchmarks and the CI matrix that re-exports fitted models from the ACEpotentials under test and requires the evaluator to reproduce them. (c) **pr/element-embeddings**: everything in this report — `ace_embedding_model`, the PCA reduction with its rank check, the factorised radial export, the distillation pipeline, the spikes and the findings.
5. **After the radial fix, revisit the recursive (DAG) AA products for the CPU evaluator**. The Phase-8 spike (`acejax/spike_recursive/`) measured subproduct sharing at **0.99×** on energy+forces for the JAX GPU evaluator in f64 (1.19× in f32; the reverse pass must materialise $\approx 25\,000$ intermediate columns, and AA is only 18% of `site_basis` on the GPU once measured against an oracle rather than in isolation — so it is a dead end there), but **2.30×** on CPU, where AA is 75% of the energy+force cost. With the radial splines no longer dominant in Julia, the AA products become its largest stage, which is the CPU regime the spike measured; `EquivariantTensors` already provides `SparseSymmProdDAG`, so the trial is a swap of the tensor’s `aabasis` on the fixed kernels.
6. **Retire the design-matrix cache** (it exhausted the storage quota at 200 GB; assembly is now minutes) and **replace `svd(R)`** in the factor-once solver with a Cholesky per λ or a threaded SVD — at $n = 46685$ it is the remaining post-assembly cost.
7. **Generator before basis**. Bounding the perturbations halved the error; adding short-range order and the unary/binary structures of GRACE’s “extended distillation” are the next data-side steps and cost minutes of GPU.
8. **Next accuracy lever, convexly**: fixed $\sqrt{\rho}$ Finnis–Sinclair columns with learned (VarPro) species weights inside the density gave −12% at degree 4 and transfer frozen to a disjoint split; the shape of the embedding function beyond $\sqrt{\cdot}$ does not matter. This is the like-for-like answer to GRACE-FS and should be included in any $S > 5$ run.

Sources: `acejax/bench/distil/RESULTS.md`; `docs/findings/FINDINGS_assembly_profile.md`, `FINDINGS_forward_profile.md`, `FINDINGS_fs_spike.md`, `FINDINGS_varpro_spike.md`, `FINDINGS_codes_survey.md`; `plan docs/plans/jax_ace_port_plan.md` (Stage 1D–E, 2B–2C).